

Improved Medicinal Plant Leaf Classification using Transfer Learning and Model Averaging Ensemble of Deep Convolutional Neural Networks

Epie F. Custodio
College of Computer Studies
Mindoro State University
Calapan City, Philippines
epiefcustodio@gmail.com

Abstract—Identification of medicinal plants is a difficult undertaking. It needs an expert's presence to recognize them. The goal of this study is to enhance existing classification and recognition accuracies of medicinal plant leaf. A model averaging ensemble of five (5) convolutional neural networks namely: ResNet50, ResNet101, ResNet152, VGG16 and VGG19 was developed which resulted in 98.22% overall accuracy, 98.40% precision rate, recall of 98.13% and 98.25% F1-score. Furthermore, transfer learning was used in the training of each network. This is done to lessen the training period of each CNN. Additionally, actual image count was increased to add variance to the data and augment training accuracy. According to the study's findings, categorizing and identifying medicinal plants using model averaging ensemble is a viable attempt in boosting overall recognition accuracy rate.

Keywords—ensemble learning, medicinal plant leaf recognition, deep learning, transfer learning, ResNet, VGGNets

I. INTRODUCTION

Over the past years, medicinal plants have been an important resource when it comes to traditional medicine. Despite the advancements in technology, traditional medicines are still being used around the world. Asia is a major biological resource[1] and the common pharmaceutical trades are from Asian medicinal plants[2]. Determining medicinal plants is a challenging task. The presence of an expert is necessary to identify them. Based on a study, about 28% of plants are medicinal in nature. Around 5% of the population in developed countries and 80% in developing ones are using medicinal plants to cure their illnesses[3]. The reason is due to government's limited facilities and its high cost of hospitalization[2].

Various studies have delved into classifying and detecting medicinal plants utilizing deep learning and transfer learning[4][5][6][7]. Some even created applications[8][9][10] to make the identification of these plants fast, accurate and automatic without the necessity of an expert in the field[11]. Achieving high accuracy is still a challenge despite the emergence of these technologies, especially in medicinal plant recognition.

Classification and detection accuracy is continuously improving. One way to improve a model's classification

accuracy is through ensemble learning[12][13][14][15][16]. Ensemble learning combines two or more models with the purpose of increasing the robustness and detection accuracy of a model[13][17].

This work's main goal is to merge deep learning neural networks including VGG16, VGG19, ResNet50, ResNet101 and ResNet152. The merging was done using average ensemble in improving classification and recognition accuracy of medicinal plant leaf. Metrics such as accuracy, recall, precision and F1-score were used in evaluating models' performance.

II. LITERATURE REVIEW

Medicinal Plant Leaf Recognition and Deep Learning

A transfer learning technique used MobileNetV2 CNN architecture on medical plant dataset, composed of 30 classes totaling to 3000 images[4]. It achieved a 98.05 percent accuracy on a held-out test set. Additionally, a training process on 34123 images and experimental results on 3570 images demonstrated the feasibility and effectiveness of medicinal plant classification, achieving a 71.3% accuracy rate from 10 classes[5].

Utilizing VGG16, a study successfully classified seven medicinal plant classes using transfer learning which includes over 7000 leaves. It achieved a 98% accuracy[18]. Furthermore, Mask R-CNN was utilized in a study. It achieved an average accuracy of 95.7% in identifying thirty medicinal plants from the Mendeley Dataset[6]. Also, different deep learning architectures on three standard datasets were used to classify medicinal plants. The results showed that VGG 16 architecture with logistic regression provides higher accuracy rate with an accuracy rate of 97.14%[19].

Using DLNN, an efficient and automatic classification system for recognizing Malaysian herbs was developed. It employed two classifiers, the Support Vector Machine (SVM) and Deep Learning Neural Network (DLNN) which achieved 74.63% and 93% recognition accuracies, respectively[9]. On the other hand, an automated medicinal plant identification system was proposed. It utilized the DeepHerb dataset comprised of 2515 leaf images from forty (40) classes. The system uses neural network techniques and deep learning to extract features and classify, outperforming pre-trained models

by 97.5%. The model is integrated into a mobile application named “HerbSnap”, that predicts herb images one(1) second per image[10].

A model was built[11] which achieves an average accuracy greater than 97% for the VNPlant-200 dataset. Common information was merged into the model that made the precision and recall better. Thus, improves identification reliability and reduces false positive rates. Similarly, a study[20] compares convolutional neural network and pretrained models VGG-16 and VGG-19 for leaf identification using a dataset of 64 classes of medicinal plants. CNN achieved an accuracy of 95.79%.

Another research[21] focused on developing a computer vision system using Convolutional Neural Networks (CNNs) and Log-Gabor filters in identifying medicinal plant leaf based on its textural features. It achieved a 98% accuracy using the DenseNet201 architecture.

In an attempt to boost recognition accuracy of medicinal plant leaf, [16] used ensemble learning combining MobileNetV2, ResNet50 and InceptionV3 which led to 99.66% accuracy from 30 classes. Additionally, [17] employed ensemble technique using VGG16, ResNet50, DensNet201, InceptionV3 and Xception leading to 98% and 99% recognition accuracies for soft ensemble and hard ensemble, respectively.

VGG Network

VGG16 is a 16-layer CNN. It contains three fully connected layers with thirteen (13) convolutional layers. The input image to the network is 224 x 224. The first two are fully connected composed of forty thousand ninety six (4096) channels while the third layer has 1000[22].

The VGG19 is a convolutional neural network designed by the University of Oxford[23] in 2014. Its architecture uses a deeper layer and smaller filters for image classification[24].

ResNet Models

ResNet-50 is composed of fifty(50) layers. This model categorizes images into 1000 classes using 224 by 224 pixels input. It undergoes five stages with convolution and identity components, offering over 23 million trainable parameters.

ResNet-101 is a improved version of ResNet50 with a 101-layer Residual Network. Convolutional layers make up the whole of the ResNet-101 in a typical configuration. Of the 33 levels, twenty-nine (29) of these blocks, use the prior block's output straight. The input for the following blocks, which were provided as residual connections, was obtained by applying the summing operator at the end of each block[25].

The 3x3 convolutional windows are used by all convolutional layers in ResNet models, and the number of filters increases with network depth.

III. METHODS

A. Dataset

The dataset is composed of 40 classes of medicinal plants found in the Philippines and is available online via Mendeley data[26]. Data description of the dataset can be found

in a study[27] utilizing the same dataset. An entirety of four thousand nine hundred twenty-two images were used in the study. The dataset was split into three sets namely: training (70%), validation (20%), and testing (10%). Fig. 1. shows the images of the different classes used in the study.

B. Environmental Setup

All the training and testing methods were carried out in the Google Colab cloud server. All experiments were run in Python v3.10.12. The backend used was TensorFlow v2.15.0 and the Application Programming Interface (API) is the Keras framework v2.15.0.

C. Data Augmentation

CNN demands substantial data for enhanced performance and excellent recognition accuracy. Overfitting usually results from not enough images in the training set. To avoid this scenario, data augmentation is necessary[28]. This can be achieved via artificial methods[29] like size scaling, horizontal shifts, and image zooming. The training set in this study was expanded on the fly. It means a new augmented dataset is generated on each iteration, which exposes the model to constantly changing augmented data variations. This augmentation process is ideal compared to traditional augmentation where dataset is single and static.

D. Transfer Learning

Establishing a new CNN from scratch is a challenging task. Utilizing a previously trained network could be an option for feature extraction using transfer learning. Transfer learning is a machine learning technique that facilitates the learning of a new task by repurposing learning from one realm to another[14]. Transfer learning was employed in this study using the five (5) CNNs: VGG16, VGG19, ResNet50, ResNet101 and ResNet152.

E. Fine Tuning and Training the Model

Hyperparameters employed to perform the investigation are listed in Table I. VGG16, VGG19, ResNet50, ResNet101, and ResNet152 were the five pre-trained networks utilized. The Adam optimizer was selected as the optimizer. The starting epoch count for each CNN selected was set to 60. The hyperparameters of each deep learning architecture were carefully adjusted by testing with several models' hyperparameter combinations and choosing the model that worked the best. Several experiments were conducted before deciding on L2 regularization and initial learning rates (LR). Batch size of 16 and a specific dropout for each CNN selected was utilized. To increase training accuracy, the last layer of each network was also unfrozen.

In the training of the five CNNs, validation losses were monitored via checkpoint. If no improvement in the training losses was observed, a callback is set using the ReduceLROnPlateau function. Likewise, if no progress is observed after a fixed period of epochs, training is terminated right away.

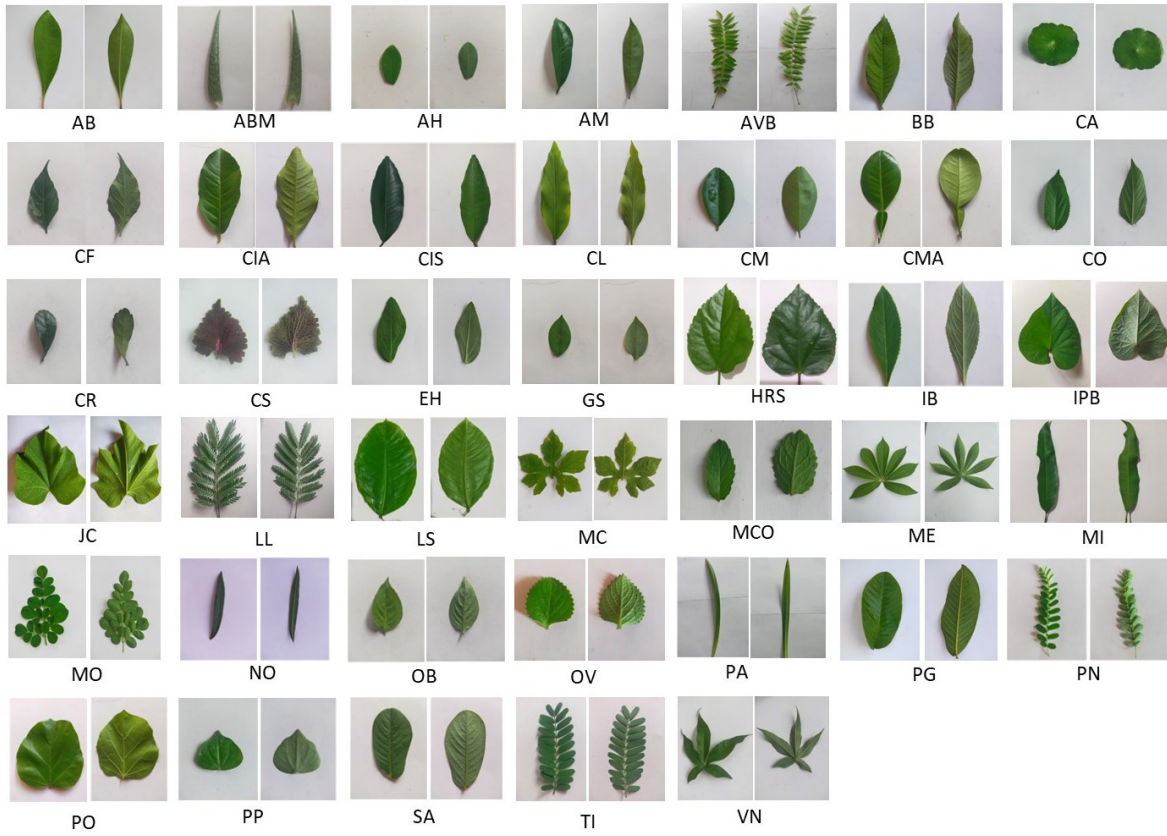


Fig. 1. Medicinal plant classification: AB-Antidesma buniis; ABM-Aloe barbadensis Miller; AH-Arachis hypogaea; AM-Annona muricata; AVB-Averrhoa bilimbi; BB-Blumea balsamifera; CA-Centella asiatica; CF-Capsicum frutescens; CIA-Citrus aurantiifolia; CIS-Citrus sinensis; CL-Curcuma longa; CM-Citrus macrocarpa; CMA-Citrus maxima; CO-Corchorus olitorius; CR-Carmona retusa; CS-Coleus scutellarioides; EH-Euphorbia hirta; GS-Gliricidia sepium; HRS-Hibiscus rosa-sinensis; IB-Impatiens balsamina; IPB-Ipomoea batatas; JC-Jatropha curcas; LL-Leucaena leucocephala; LS-Lagerstroemia speciosa; MC-Momordica charantia; MCO-Mentha cordifolia Opiz; ME-Manihot esculenta; MI-Mangifera indica; MO-Moringa oleifera; NO-Nerium oleander; OB-Ocimum basilicum; OV-Origanum vulgare; PA-Pandanus amaryllifolius; PG-Psidium guajava; PN-Phyllanthus niruri; PO-Premna odorata; PP-Pepromia pellucida; SA-Senna alata; TI-Tamarindus indica; VN-Vitex negundo. Source: [27]

TABLE I. MODELS' HYPERPARAMETER SETTINGS

CNN Model	Optimizer	Epochs	Initial Learning Rate	L2 regularizer	Batch Size	Drop Out Rate	"Freezed" Layers
VGG16	Adam	60	.00001	.0001	16	0.10	12
VGG19	Adam	60	.0001	.0001	16	0.10	12
ResNet50	Adam	60	.0001	.0001	16	0.50	75
ResNet101	Adam	60	.0001	.0001	16	0.35	10
ResNet152	Adam	60	.0001	.0001	16	0.50	12

F. Ensemble of Models

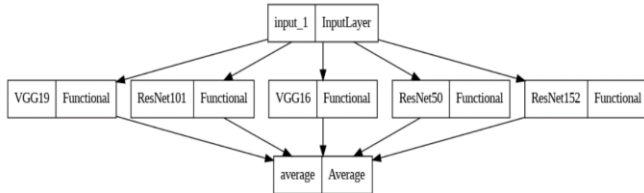


Fig. 2. Ensemble of models

Ensemble of models were tried to identify the best model for recognition and classification of medicinal plants. Seven distinct combinations were analyzed and trained, and the best combination was chosen. Fig. 2. shows the ensemble of models of the five CNNs using model averaging. Ensemble

averaging allows each distinct model to generate opinions on the final prediction.

G. Evaluation Metrics

In this study, the models were evaluated based on performance metrics including accuracy, recall, precision, and F1-Score. The formula for calculation of the metrics are shown below:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

$$F1 - Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (4)$$

IV. EXPERIMENTAL RESULTS

Shown in Fig. 3 are the training and validation accuracies of each network while Fig. 4 shows the training and validation losses. The number of epochs for each training was set to 60 and corresponding learning rates were also manually set. After several trials, the best learning rate was selected. The L2 regularizers were also set to determine the best model. After several epochs with no improvement in the accuracy, the learning rate is increased and eventually halted once the designated number of epochs was reached. Of the five CNNs, ResNet50 had the best convergence as demonstrated in Fig. 3a.

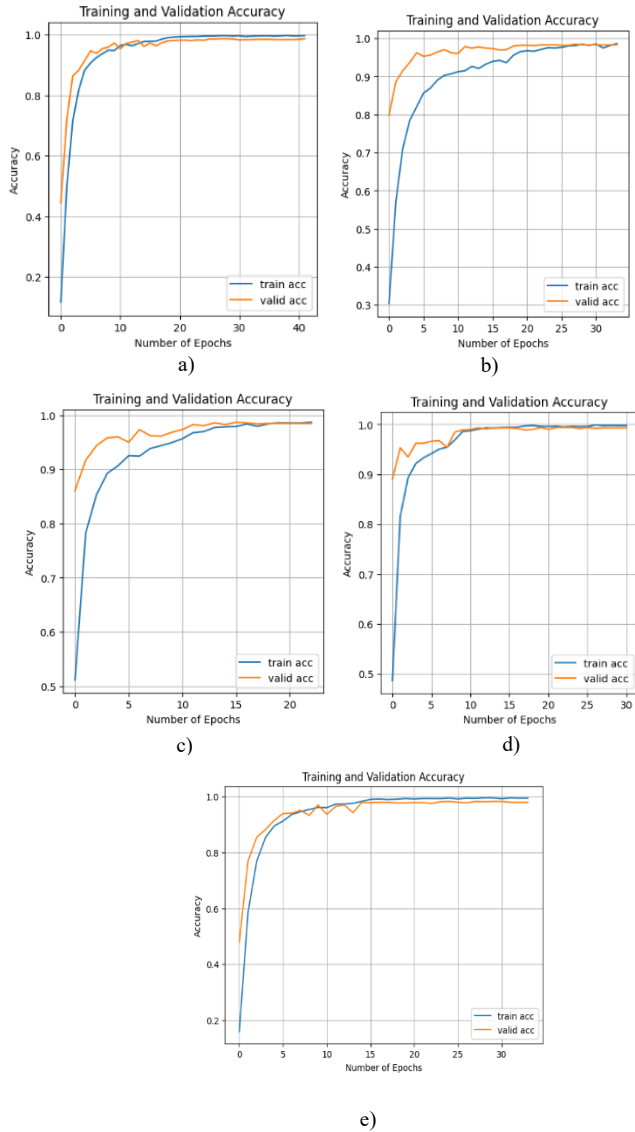


Fig. 3. Training and Validation Accuracy of 5 CNNs: (a) ResNet50; (b) ResNet101; (c) ResNet152 (d) VGG16; (e) VGG119.

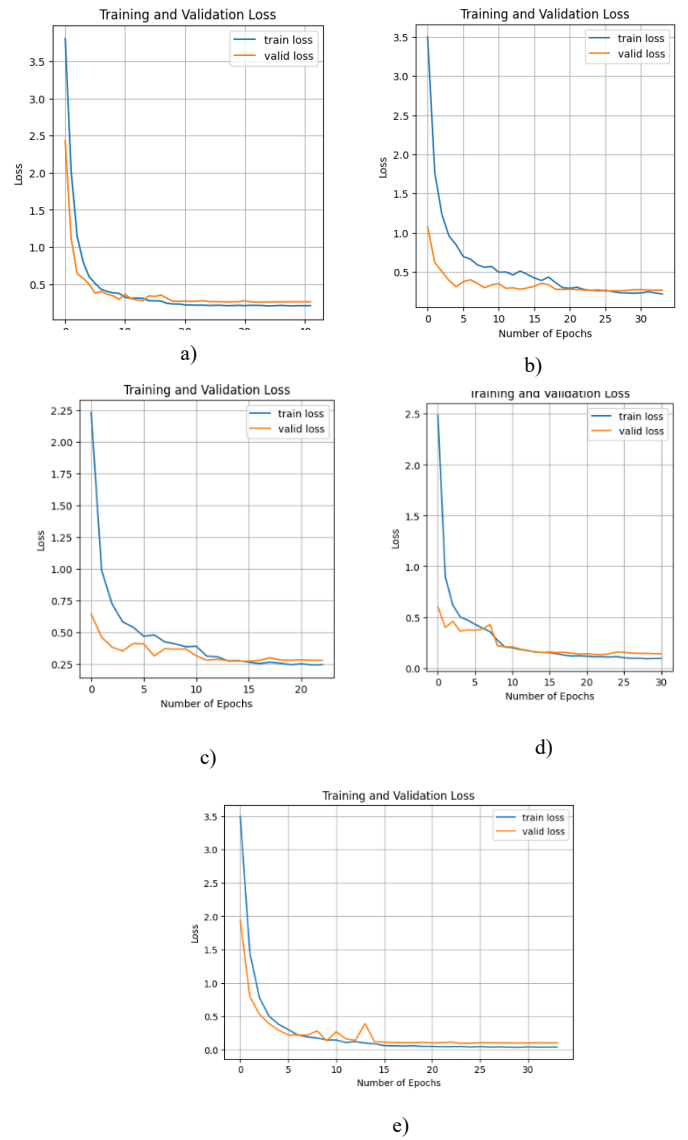


Fig. 4. Training and Validation Loss: (a) ResNet50; (b) ResNet101; (c) ResNet152; (d) VGG16; (e) VGG119.

TABLE II. PERFORMANCE ACCURACIES OF INDIVIDUAL CNN MODEL

CNN Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
VGG16	94.46	94.93	94.23	94.30
VGG19	90.89	93.08	91.15	91.00
ResNet50	95.25	96.12	95.26	95.20
ResNet101	91.68	93.42	91.42	91.60
ResNet152	92.08	93.50	92.20	92.22

Table II shows the performance accuracies of each individual CNN including precision, recall and F1-score. ResNet50 garnered the highest accuracy resulting to 95.25%, precision of 96.12%, recall of 95.26% and F1-score of 95.20%. Next to the highest is VGG16 with 94.46% accuracy result, 94.93% precision, 94.23% recall and 94.30% F1-score. Additionally, ResNet152 came third with results of 92.08%, 93.50%, 92.20% and 92.22% for accuracy, precision, recall and

F1-score, respectively. ResNet101 is next with 91.68% accuracy result, 93.42% precision, 91.42% recall and 91.60% F1-score. Lastly, VGG19 had the lowest accuracy among the five CNNs investigated, with the following results: 90.89% accuracy, 93.08% precision, 91.15% recall and 91% F1-score.

TABLE III. ENSEMBLE OF MODELS

Ensemble	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
VGG19 + VGG16	95.84	96.23	95.78	95.75
Resnet50 + Resnet152 + Resnet101	97.82	98.05	97.85	98.85
VGG19 + VGG16 + Resnet50	96.83	97.05	96.80	96.78
Resnet50 + Resnet152 + VGG19	96.44	96.58	96.18	96.25
Resnet50 + Resnet152 + VGG16	97.23	97.28	97.13	97.10
Resnet50 + Resnet152 + Resnet101 + VGG16	97.43	97.63	97.35	97.30
Resnet50 + Resnet152 + Resnet101 + VGG16 + VGG19	98.22	98.40	98.13	98.25

Table III reveals the ensemble of models. Ensemble of VGG19 and VGG16 resulted to 95.84% accuracy. Whereas ensemble of the three (3) ResNet models lead to a 97.82% accuracy. The combination of VGG Nets and ResNet50 resulted to a 96.83% accuracy. Moreover, Resnet50 + Resnet152 + VGG19 revealed a 96.44%. On the other hand, Resnet50 + Resnet152 + VGG showed 97.23% accuracy. Amalgamation of the ResNet models and VGG16 gave an accuracy of 97.43%. Finally, the ensemble of the five (5) CNNs resulted in a 98.22% accuracy which established the highest accuracy rate. The result of the ensemble of models in Table III found to outperform individual classification accuracies of each model in Table II.

Table IV presents existing studies focusing on classifying medicinal plant leaf diseases. Overall classification and recognition accuracies of each model in the studies presented were given in the table. The utilization of ensemble learning boosted the overall accuracy of the model as compared to previous studies that did not use ensemble learning. With the fusion of models via ensemble learning, each model has the equal chance to give its prediction and added to the final prediction. With this, it reveals that average ensemble can enhance recognition accuracy of medicinal plants. Instead of using just one model, two or more models are combined to do the predictions.

The ensemble learning strategy yields a greater accuracy rate than other methods, as Table IV illustrates. Though the current study did not outperform previous studies that utilized ensemble learning, the result is slightly similar. Other factors such as number of classes and number of images may be a consideration aside from the resulting accuracy of the ensembled model.

TABLE IV. EXISTING MODELS COMPARED TO THE PROPOSED MODEL

Study	Dataset	Number of Images	Number of classes	Method Used	Accuracy (%)
[18]	-	7000	7	VGG-16	98
[4]	-	3000	30	MobileNetv2	98.05
[5]	-	34123	10	CNN	71.3
[6]	-	1800	30	Mask R-CNN	95.7
[19]	Leaf12	320	12	VGG19 + LR (Linear Regression)	97.14
[30]	Own Dataset	6,628	25	ResNeXt-50	97.68
[9]	Own Dataset	1000	20	DLNN	93
[10]	DeepHerb	2515	40	Xception + ANN	97.5
[11]	VNPlant-200	20000	200	Xception	97
[20]	Own Dataset	64000	64	VGG16, VGG19	97.8, 97.6
[21]	Centre of Plant Medicine Research, Ghana (MyDataset)	2450	49	OTAMNet (hybrid of DenseNet 201 and Log Gabor)	98
[17]	Own Dataset	5000	10	Ensemble (VGG16, ResNet50, DensNet201, Inception V3, Xception)	Soft ensemble-98% Hard ensemble-99%
[16]	Mendeley Dataset	1841	30	Ensemble (MobileNetV2, ResNet50, Inception V3)	99.6%
Own Study	PhilMedic (Own Dataset)	4922	40	Ensemble (ResNet50, ResNet101, ResNet152, VGG16, VGG19)	98.22

VI. CONCLUSIONS

Accurate identification of medicinal plants is imperative. It will help common people in treating their diseases without worries of misclassification or possible side effects. With artificial intelligence, medicinal plant leaf identification is enhanced further. The development of an application that automatically detects medicinal plant leaf without the need of an expert is a big help to the community.

Several attempts have been made already to detect medicinal plant leaf using transfer learning and deep learning.

To augment the recognition accuracy, this study used average ensemble learning. Five transfer learning networks were utilized namely: RestNet50, RestNet101, ResNet152, VGG16 and VGG19. Of the CNNs used, seven different combinations were tried and analyzed. The amalgamation of 5 networks was found to have the highest accuracy. However, the limitation of the study is its shortage of images available for training. Therefore, data augmentation on the fly was devised to enhance and increase image count.

For future studies, detection of medicinal plants may include other plant parts in the dataset aside from the leaf which may include barks, stems, flowers and others. Other ensemble techniques might also be experimented aside from average ensemble which may further increase accuracy.

Overall, the result of the study showed that it is feasible to classify and identify medicinal plant leaf applying model averaging ensemble with increased accuracy.

REFERENCES

- [1] C. P. Kala, P. P. Dhyani, and B. S. Sajwan, "Developing the medicinal plants sector in northern India: Challenges and opportunities," *J. Ethnobiol. Ethnomed.*, vol. 2, no. August 2006, 2006, doi: 10.1186/1746-4269-2-32.
- [2] B. Dey, J. Ferdous, R. Ahmed, and J. Hossain, "Heliyon Assessing deep convolutional neural network models and their comparative performance for automated medicinal plant identification from leaf images," *Heliyon*, vol. 10, no. 1, p. e23655, 2024, doi: 10.1016/j.heliyon.2023.e23655.
- [3] R. Azadnia, M. M. Al-Amidi, H. Mohammadi, M. A. Cifci, A. Daryab, and E. Cavallo, "An AI Based Approach for Medicinal Plant Identification Using Deep CNN Based on Global Average Pooling," *Agronomy*, vol. 12, no. 11, p. 2723, 2022, doi: 10.3390/agronomy12112723.
- [4] J. Abdollahi, "Identification of Medicinal Plants in Ardabil Using Deep learning: Identification of Medicinal Plants using Deep learning," *Proc. - 2022 27th Int. Comput. Conf. Comput. Soc. Iran, CSICC 2022*, no. May, 2022, doi: 10.1109/CSICC55295.2022.9780493.
- [5] R. Akter and M. I. Hosen, "CNN-based Leaf Image Classification for Bangladeshi Medicinal Plant Recognition," *ETCCE 2020 - Int. Conf. Emerg. Technol. Comput. Commun. Electron.*, 2020, doi: 10.1109/ETCCE51779.2020.9350900.
- [6] L. Almazaydeh, R. Alsalamien, and K. Elleithy, "Herbal Leaf Recognition Using Mask-Region Convolutional Neural Network (Mask R-Cnn)," *J. Theor. Appl. Inf. Technol.*, vol. 100, no. 11, pp. 3664–3671, 2022.
- [7] A. Bahri, Y. Bourass, I. Badi, H. Zouaki, and K. El, "Dynamic CNN Combination for Morocco Aromatic and Medicinal Plant Dynamic CNN Combination for Morocco Aromatic and Medicinal Plant Classification," no. October, 2023, doi: 10.12785/ijcds/110120.
- [8] P. Swathika, A. Ajithar, Z. Hishaamudin, and E. Nitheeswaran, "Medi-Plant : A Deep Learning Approach for Medicinal Plant Classification with Pix2Pix Generative Adversarial Network Medi - Plant : A Deep Learning Approach for Medicinal Plant Classification with Pix2Pix Generative Adversarial Network," pp. 0–22, 2024.
- [9] A. Muneer and S. M. Fati, "Efficient and automated herbs classification approach based on shape and texture features using deep learning," *IEEE Access*, vol. 8, pp. 196747–196764, 2020, doi: 10.1109/ACCESS.2020.3034033.
- [10] S. Roopashree and J. Anitha, "DeepHerb: A Vision Based System for Medicinal Plants Using Xception Features," *IEEE Access*, vol. 9, pp. 135927–135941, 2021, doi: 10.1109/ACCESS.2021.3116207.
- [11] L. Shahmiri, P. Wong, and L. S. Dooley, "Accurate Medicinal Plant Identification in Natural Environments by Embedding Mutual Information in a Convolution Neural Network Model," *5th IEEE Int. Image Process. Appl. Syst. Conf. IPAS 2022*, 2022, doi: 10.1109/IPAS55744.2022.10053008.
- [12] N. Chowdhury, J. Sultana, T. Rahman, T. Chowdhury, and F. T. Khan, "Potato leaf disease detection through ensemble average deep learning model and classifying the disease severity," no. May, pp. 494–502, 2024, doi: 10.11591/ijeecs.v35.i1.pp494-502.
- [13] V. Anand *et al.*, "Weighted Average Ensemble Deep Learning Model for Stratification of Brain Tumor in MRI Images," *Diagnostics*, vol. 13, no. 7, 2023, doi: 10.3390/diagnostics13071320.
- [14] G. Fenu and F. M. Mallocci, "Classification of Pear Leaf Diseases Based on Ensemble Convolutional Neural Networks," no. i, pp. 141–152, 2023.
- [15] A. Ganguly, "Ensemble Learning for Plant Leaf Disease Detection : A Novel Approach for Improved Classification Accuracy," pp. 1–24, 2023.
- [16] S. Sachar and A. Kumar, "Deep ensemble learning for automatic medicinal leaf identification," *Int. J. Inf. Technol.*, vol. 14, no. 6, pp. 3089–3097, 2022, doi: 10.1007/s41870-022-01055-z.
- [17] A. H. Uddin *et al.*, "Deep-Learning-Based Classification of Bangladeshi Medicinal Plants Using Neural Ensemble Models," *Mathematics*, vol. 11, no. 16, p. 3504, 2023, doi: 10.3390/math11163504.
- [18] P. Architha Shastri and A. P. Patil, "Classification of Medicinal Leaves Using Support Vector Machine, Convolutional Neural Network and You only Look Once," *Proc. B-HTC 2020 - 1st IEEE Bangalore Humanit. Technol. Conf.*, 2020, doi: 10.1109/B-HTC50970.2020.9297878.
- [19] S. Anubha Pearline, V. Sathiesh Kumar, and S. Harini, "A study on plant recognition using conventional image processing and deep learning approaches," *J. Intell. Fuzzy Syst.*, vol. 36, no. 3, pp. 1997–2004, 2019, doi: 10.3233/JIFS-169911.
- [20] A. Paulson and S. Ravishankar, "AI Based Indigenous Medicinal Plant Identification," *Proc. - 2020 Adv. Comput. Commun. Technol. High Perform. Appl. ACCTHPA 2020*, pp. 57–63, 2020, doi: 10.1109/ACCTHPA49271.2020.9213224.
- [21] S. O. Oppong, F. Twum, J. Ben Hayfron-Acquah, and Y. M. Missah, "A Novel Computer Vision Model for Medicinal Plant Identification Using Log-Gabor Filters and Deep Learning Algorithms," *Comput. Intell. Neurosci.*, vol. 2022, 2022, doi: 10.1155/2022/1189509.
- [22] N. Izdihar, S. B. Rahayu, and K. Venkatesan, "Comparison Analysis of CXR Images in Detecting Pneumonia Using VGG16 and ResNet50 Convolution Neural Network Model," *Int. J. Informatics Vis.*, vol. 8, no. 1, pp. 326–332, 2024, doi: 10.62527/joiv.8.1.2258.
- [23] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *3rd Int. Conf. Learn. Represent. ICLR 2015 - Conf. Track Proc.*, pp. 1–14, 2015.
- [24] G. Fenu and F. M. Mallocci, "Using Multioutput Learning to Diagnose Plant Disease and Stress Severity," *Complexity*, vol. 2021, 2021, doi: 10.1155/2021/6663442.
- [25] R. Kumari and J. Wasim, "A Deep Learning Approach for Human Facial Expression Recognition using Residual Network – 101," *J. Curr. Sci. Technol.*, vol. 13, no. 3, pp. 517–532, 2023, doi: 10.59796/jest.V13N3.2023.2152.
- [26] E. F. Custodio, "PhilMedic: Philippine Medicinal Plant Leaf Dataset," 2024. <https://data.mendeley.com/datasets/tsvdyhbpbs/1>
- [27] E. F. Custodio, "Classifying Philippine Medicinal Plants Based on Their Leaves Using Deep Learning," *2023 IEEE World AI IoT Congr. AIIoT 2023*, pp. 29–35, 2023, doi: 10.1109/AIIoT58121.2023.10174335.
- [28] B. Saiharsha, A. Abel Leslie, B. Diwakar, R. Karthika, and M. Ganesan, "Evaluating performance of deep learning architectures for image classification," *Proc. 5th Int. Conf. Commun. Electron. Syst. ICCES 2020*, no. Icces, pp. 917–922, 2020, doi: 10.1109/ICCES48766.2020.09137884.
- [29] C. Shorten and T. M. Khoshgoftaar, "A survey on Image Data Augmentation for Deep Learning," *J. Big Data*, vol. 6, no. 1, 2019, doi: 10.1186/s40537-019-0197-0.
- [30] G. Kumar and V. Kumar, "Herbal Plants Leaf Image Classification Using Deep Learning Models Based On Augmentation Approach," *SSRN Electron. J.*, 2022, doi: 10.2139/ssrn.4292344.